PocketGigaScan

Panorama Stitching Feature Manual

Import JPEG originals from one panorama, check the grid and registration, choose an output format, stitch locally and inspect the result. Photos are copied into an app-owned task directory; stitching and export run on this device.



Main workspace example with a synthetic texture. The installed version may differ.

English edition | Based on the current application interface

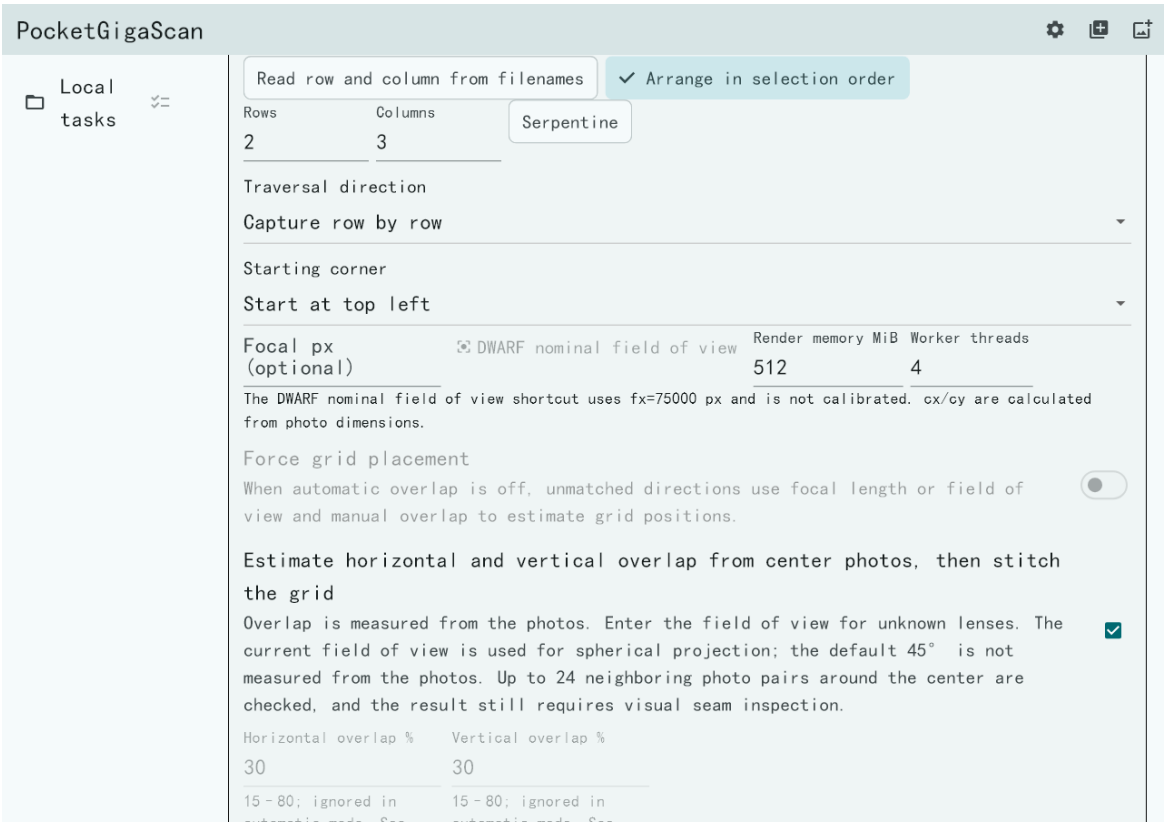
Import Photos and Check Their Arrangement

Select Import photos in the main window, then choose .jpg/.jpeg photos from one panorama. After task creation, expand Grid and camera and check each photo's position.

1. If filenames contain recognizable row and column information, start with Read row and column from filenames. Otherwise choose Arrange in selection order.
2. For manual arrangement, enter rows and columns, then choose row-wise or column-wise traversal, the starting corner and serpentine order.
3. Review thumbnails and coordinates across the grid. Correct any ordering or row/column errors before stitching.
4. Clicking a photo selects it; use an explicit lock action to pin its grid position.

Accuracy boundary

Standard single-task import accepts JPEG/JPG only. Filename inference and camera traversal may be wrong; forced grid placement is an estimate and needs review. Originals are copied to a local task directory and are not uploaded by importing.



Grid and camera settings with the photo thumbnails.

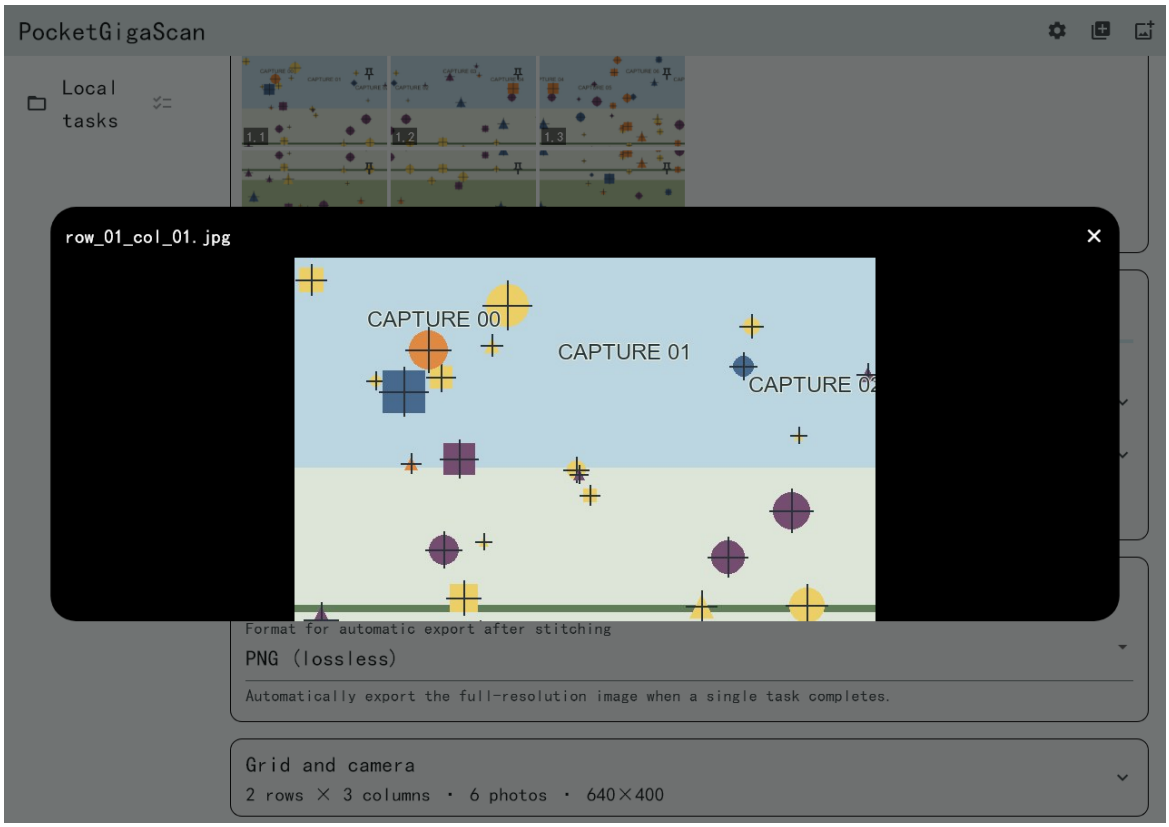
Double-Click Preview and Registration Diagnostics

Double-click a grid photo to preview the original. Previewing does not change photo selection or grid lock state. The task page also lists neighboring-photo registration issues to review.

1. Double-click a photo in the grid to open the preview on a dark background.
2. Pan or zoom to inspect details, then select the close button to return to the grid.
3. Expand registration issues and select an entry to inspect the related neighboring photos.
4. Check grid mapping, texture and overlap before deciding whether to correct or lock a photo.

Preview limits

The preview uses size-limited decoding, capped at a 2,048-pixel edge. It is for viewing the source photo, not full-resolution editing. Low texture, parallax, moving objects or no clean neighboring coverage can affect automatic registration.



Double-click a grid photo to preview its source.

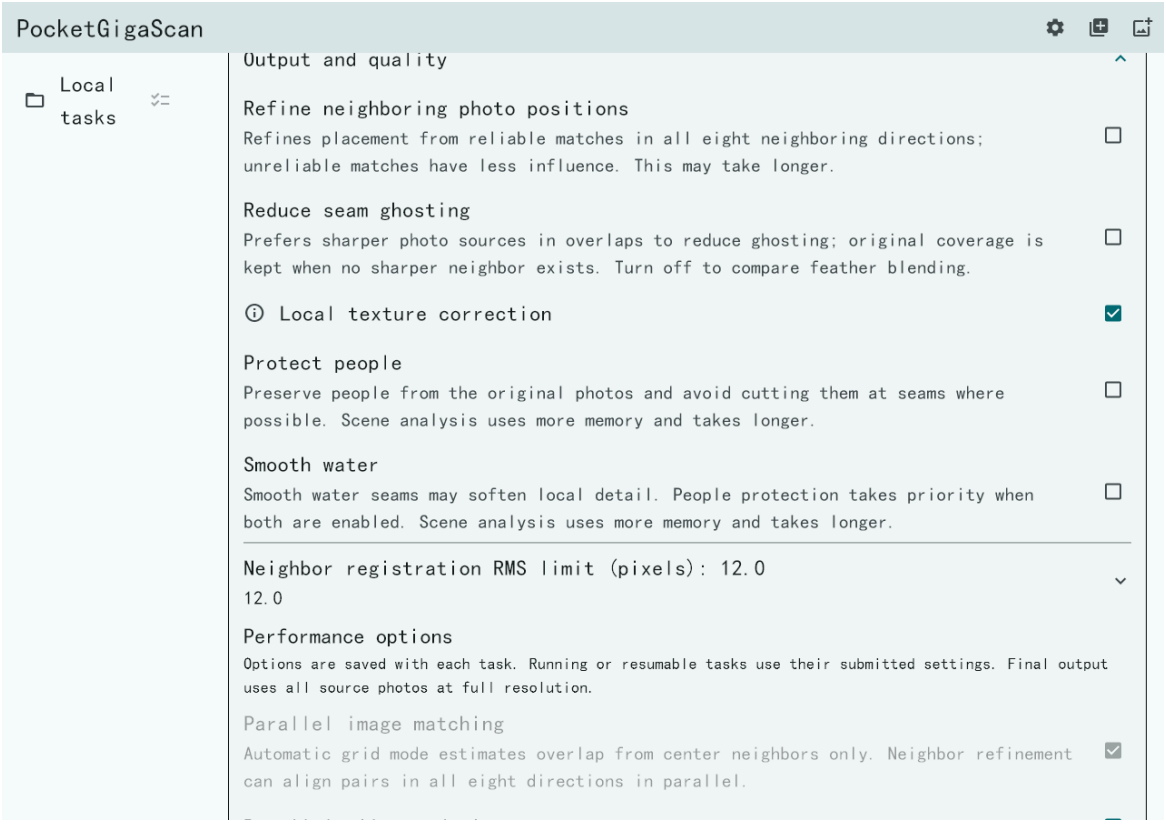
Choose Output Format and Quality

Before stitching, expand Output and quality and choose a panorama format for the current task. Initial defaults are JPEG on Windows/macOS, TIFF on Android and PNG on iOS; saved tasks keep their own processing settings.

1. Choose PNG, TIFF/BigTIFF, JPEG XL or JPEG according to how the image will be used.
2. Review seam blending and local texture warp; keep the defaults if unsure.
3. People protection and water smoothing are available on Windows/macOS; both are independent and off by default.
4. Confirm the format, output location and task settings before starting.

JPEG and JPEG XL

JPEG is lossy at fixed quality 90 and flattens transparency onto white. If either edge exceeds 65,500 pixels, the image is proportionally reduced. Consider TIFF/BigTIFF for a larger edge. JPEG XL is lossy only and requires the app to load libjxl.



Panorama output format and quality options.

Run a Task and Read Its Status

The task status card shows the stitching stage and progress. Stitch details and Stitching log are separate panels for task/result information and processing steps.

1. Confirm the task is ready, then start stitching. Large grids may require significant time and memory.
2. Use pause, resume or cancel when available. A mobile app sent to the background may request a cooperative pause.
3. Expand Stitch details for output format and diagnostics; expand Stitching log for timestamps and processing stages.
4. If an error occurs, read its summary and technical details, then check photo arrangement, storage and output folder.

Timing information

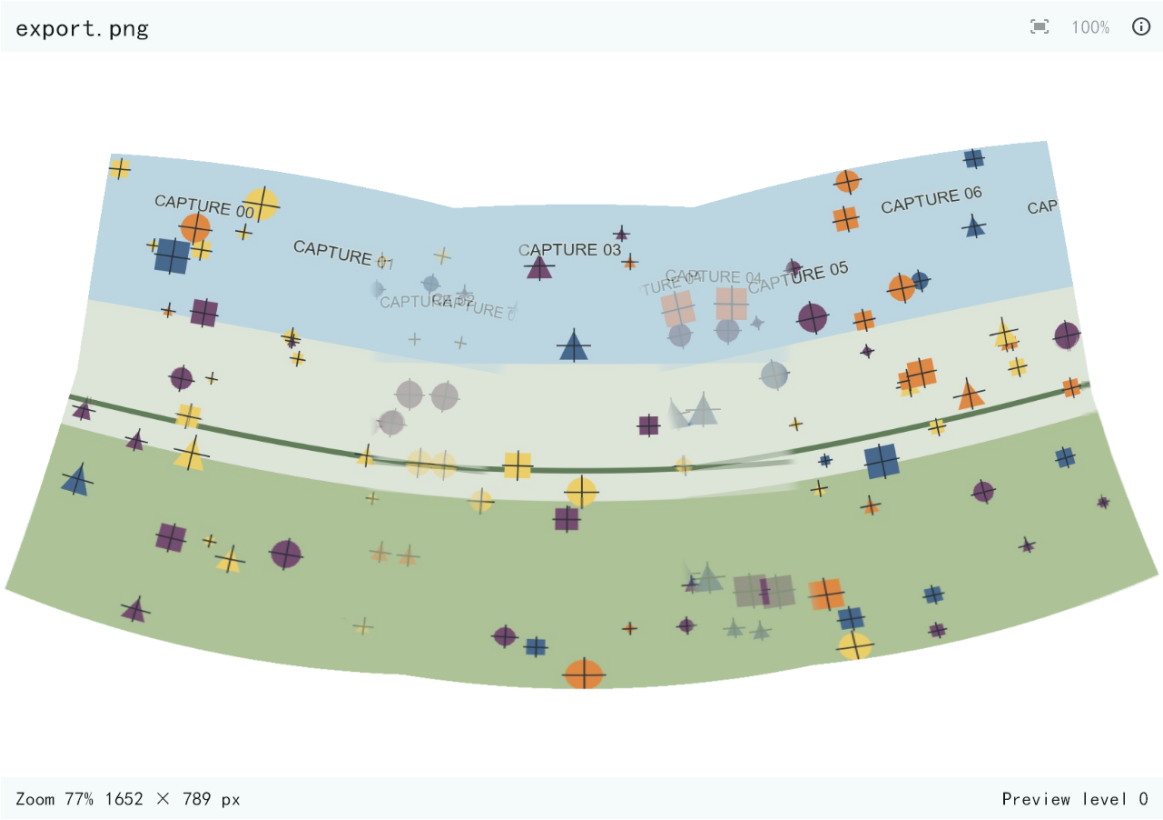
Recent tasks include stage timing; older tasks may show unknown time. Open the viewer to inspect automatic stitching results.

View a Panorama and Inspect Sources

Open a completed panorama in the tiled viewer and use zoom and drag to inspect seams, edges and details. Mobile devices support touch gestures.

1. Open the exported image viewer and fit the panorama to the window.
2. Zoom into seams and areas with possible movement, then drag to inspect other locations.
3. Select Inspect pixel source and click a location to view photos that geometrically cover it.
4. Use Show image in folder on desktop or the save/share action on mobile to access the export.

> **Source inspection is not blend tracing**
>
> The source panel lists geometrically covering photos and neighbor diagnostics. It does not infer which photo supplied the renderer's final blended pixels. Some obstructions have no clean neighboring coverage and cannot be guaranteed to disappear.



Native-rendered synthetic example; 4 of 6 tile positions are grid-estimated and need visual review. This is not image-quality acceptance.

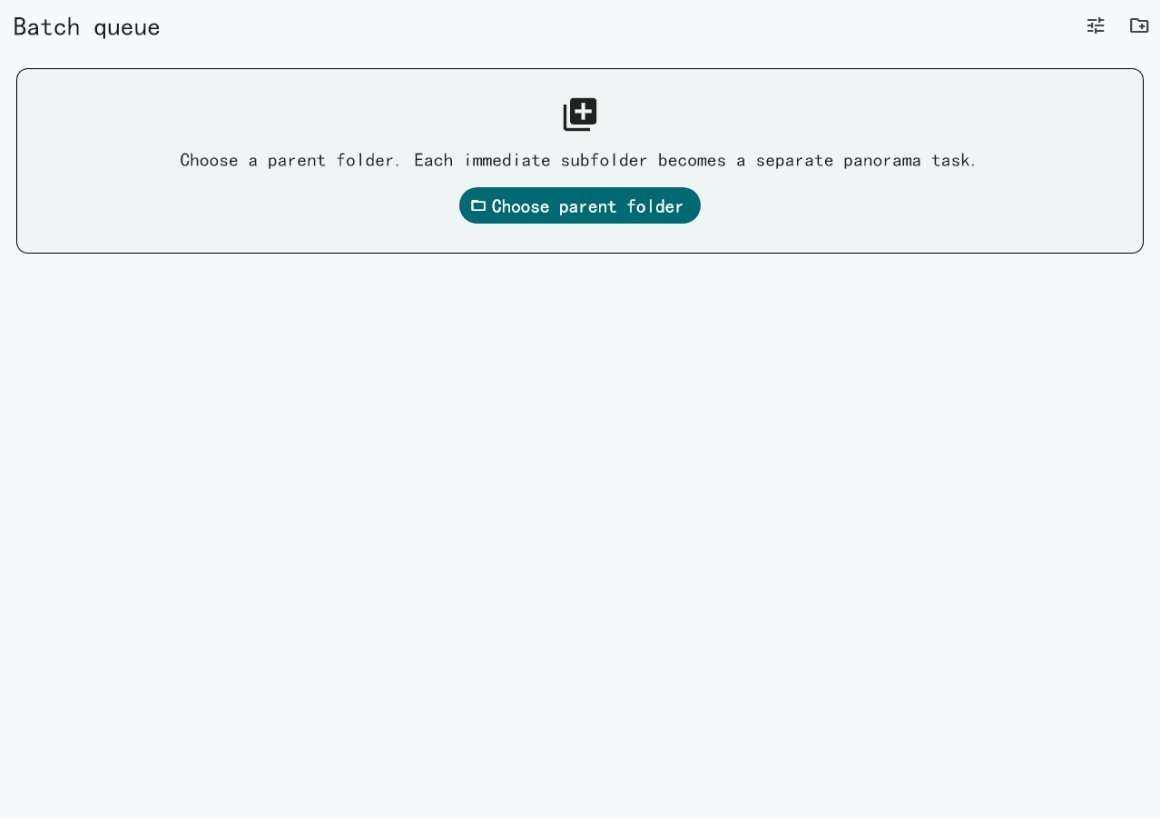
Process Folders in a Batch

The batch queue supports workflows where each panorama is stored separately: choose a parent folder and each direct child folder becomes an independent panorama task.

1. Open Batch queue and choose a parent folder containing panorama subfolders.
2. Confirm that each child folder contains originals for one panorama only.
3. Edit rows, columns and field of view for a task if needed, then start the queue.
4. Check each task's status in the queue and add another batch when needed.

Processing scope

The batch workflow uses direct child folders; it does not promise recursive scanning at arbitrary depth. Concurrency is bounded by the device resource policy.



Batch queue entry point before selecting a parent folder.

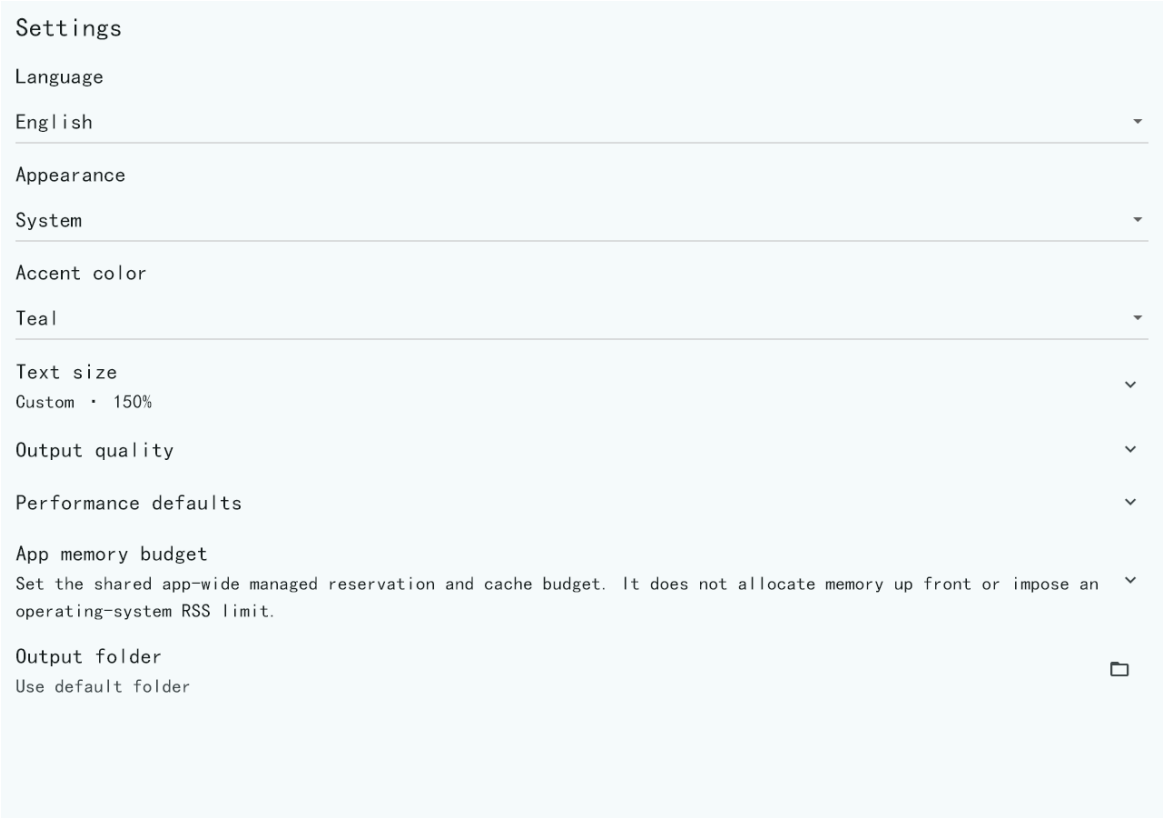
Set Output and Performance Defaults

The gear button opens Settings for language, appearance, accent color, output folder, quality and performance defaults. Defaults apply to new tasks; existing tasks retain their saved options.

1. Choose system, Chinese or English; select light/dark appearance and an accent color.
2. Choose a common output format and default output folder. Windows/macOS write directly to the selected folder.
3. Review performance defaults and the shared memory budget. Keep automatic or recommended settings if unsure.
4. Android keeps a private result and copies it to a persistent output folder; a failed copy can be retried.

Changing a completed task

Completed tasks hide processing settings. Create a copy before changing options and stitching again. Removing a task removes its record and queue reference but keeps the originals and exported images.



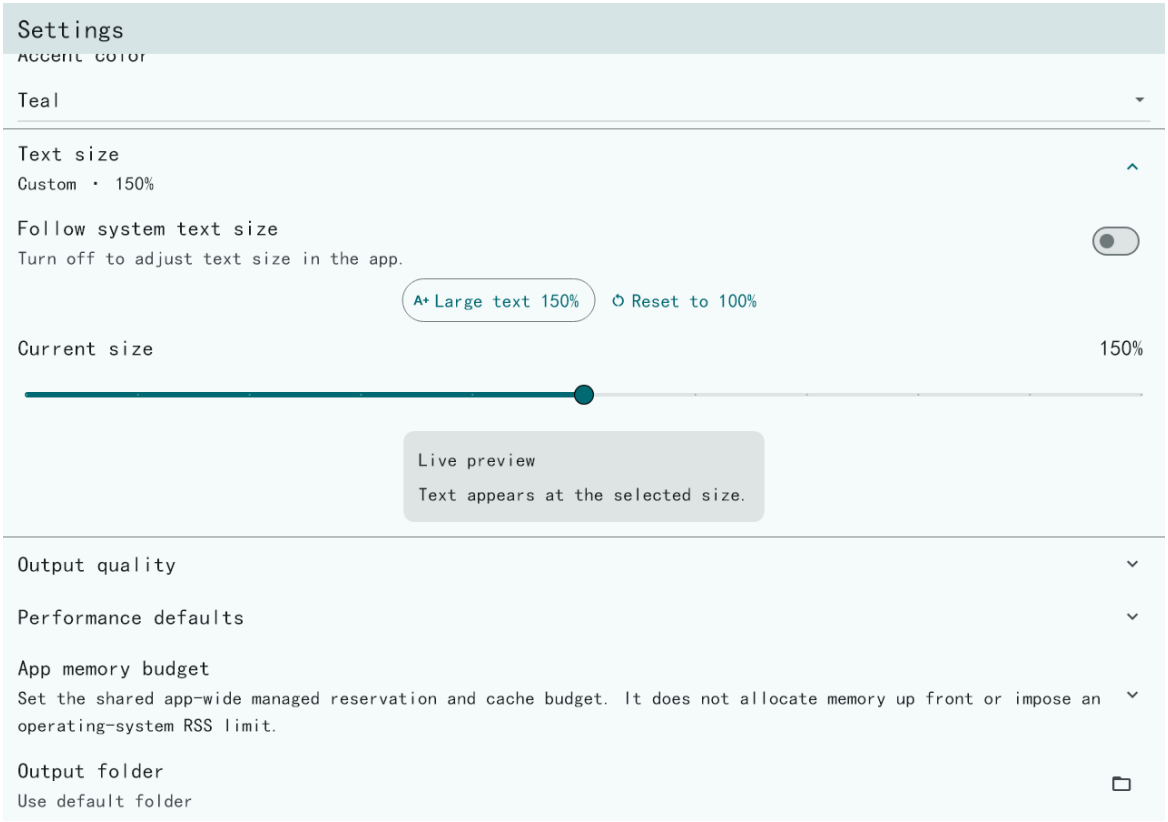
Application language, output and default settings.

Adjust Interface Text Size

Use Text size in Settings to choose how the interface scales text. Follow system text size is on by default, so the app follows the device's system text scale.

1. Leave Follow system text size on to use the current system text scale.
2. Turn it off to set text size manually from 100% to 200% in 10% steps.
3. Select Large text 150% for a quick 150% setting, or Reset to 100% to return to the standard scale.
4. Review the live preview, then check text on task, settings and viewer screens.

> **Older settings**
>
> Configurations without a saved text-size option default to following the system text scale. Long labels may wrap at larger scales; content areas can scroll.



Text-size controls, quick adjustments and live preview.

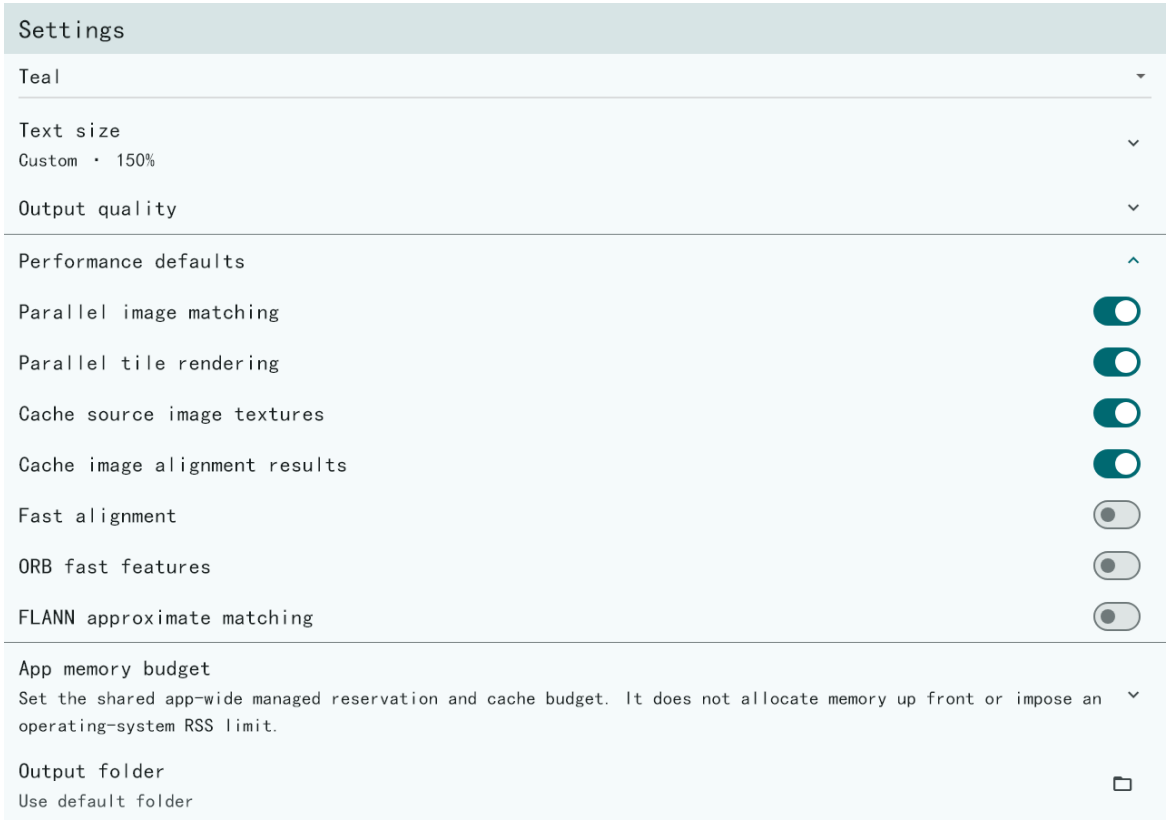
Performance Options and Mobile DWARF3

Desktop users can adjust matching, rendering, caches and memory options. Mobile also provides DWARF3 panorama downloads. Both features depend on device and network conditions.

1. Change parallel, cache or fast-registration defaults only when you understand the resource and quality tradeoffs.
2. On mobile, connect to a local Wi-Fi network that can reach the camera, then open the DWARF3 panoramas page.
3. Check the device address (default 192.168.88.1), connect or refresh the list, select panoramas and download.
4. Open a downloaded task, check its arrangement and continue with local stitching.

Platform boundary

People protection and water smoothing are Windows/macOS-only, use an offline bundled model and are hidden on phones. DWARF3 downloads originals; it does not remotely capture or control the camera. Verify connectivity on the target network and device.



Performance defaults; DWARF3 import is available on mobile.

Format Comparison and Common Issues

Choose a format based on compatibility, file size and whether you need lossless editing. Actual output is also limited by device resources and encoder constraints.

Review the result

Automatic arrangement, grid placement and overlap rates may be estimates. Movement, parallax and no clean neighboring coverage can leave visible artifacts; zoom in to inspect the actual result.

Format	Compression	Useful for	Notes
PNG	Lossless	General viewing and editing	Can be large; preserves transparent pixels.
TIFF / BigTIFF	Lossless	High fidelity and large files	Target software must support the TIFF variant.
JPEG XL	Lossy	Higher compression efficiency	Requires libjxl to be loaded; viewer compatibility varies.
JPEG	Lossy quality 90	Broad sharing	White transparency; 65,500-pixel edge limit, then reduced.

Issue	Check first
Order is wrong	Grid mode, row/column counts, starting corner and traversal direction.
Photos do not register	Clear texture and overlap between neighbors; inspect flagged photos.
Ghosting at a seam	Moving objects or parallax. Review desktop seam options and the result; full repair is not guaranteed.
Export does not finish	Disk space, folder permissions and selected format; Android can retry output copying.
An image is missing after task removal	Removing a task keeps originals and exports. Check their retained locations.